

Course: MCA

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01.

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[illegible]

• Assignment No 1.

Discrete Frequency Distribution.

1. In a Survey of 20 families, the following data regarding the number of children in a family was collected. Prepare a discrete frequency distribution for the following Data.
Data:- 1, 0, 1, 3, 4, 2, 2, 3, 4, 1, 0, 4, 2, 3, 3, 5, 3, 1, 3, 1

No. of children	Count	No. of Families.
0	II	2
1	IIII	5
2	III	3
3	IIII 1	6
4	III	3
5	I	1
		Total = 20.

2. Consider 40 Families in Village and number of children per family was recorded and following data is obtained. Prepare a discrete frequency distribution for the following data.

Data:-

1 0 0 2 1 5 6 2 2 1 0 3
4 2 1 6 3 2 1 5 3 3 2 4
2 3 3 0 2 1 4 5 3 3 4 4
1 2 4 5

No. of children	Count	No. of Families
0		4
1		7
2		10
3		7
4		6
5		4
6		2
		Total = 40

Q.3] Following are the marks obtained by 25 students in a test. Prepare a discrete frequency distribution table.
Data:-

15, 35, 20, 20, 30, 25, 25, 10, 15, 35, 40, 50, 45, 10, 25, 20, 35, 30, 40, 45, 40, 35, 20, 25, 20.

Marks	Count	No. of Students
10		2
15		2
20		5
25		4
30		2
35		4
40		3
45		2
50		1
		Total = 25

Continuous Frequency Distribution

Q.1] Prepare a frequency distribution table using inclusive class method. Prepare a frequency table by taking class interval of 110-114, 115-119, 120-124, 125-129, 130-134, 135-139, 140-144. Total: 36.

125 125 112 128 133 128 135 134 131
135 122 115 129 136 129 116 121 135
131 125 137 124 126 130 127 134 129
122 122 115 120 121 132 120 135 112

class	Frequency	Count
110-114		2
115-119		3
120-124		4
125-129		5
130-134		5
135-139		4
140-144	-	0
Total		36

Q.2] Prepare a frequency distribution table using inclusive class method. Prepare a frequency table by taking class interval of 10-13, 14-17, 18-21 - Exclusive class method.

Total = 40.

Date: 10 11 29 32 23 15 18 33 10 29
17 16 18 14 27 18 15 38 16 19
15 19 25 17 30 24 22 34 20 23
22 24 20 12 36 28 13 22 31 26.

Inclusive class method.

→	class	Frequency	Count
	10-13		5
	14-17		8
	18-21		8
	22-25		7
26-29	26-28		5
30-33	29-31		4
34-37	32-34		2
38-41	35-37		1
	38-40		
			Total = 40

Exclusive class method.

$$14-13=1 \rightarrow 1/2 = 0.5$$

Lower limit - 0.5.

Upper limit + 0.5.

class.	Frequency	Count.
9.5-13.5		5
13.5-17.5		8
17.5-21.5		8
21.5-25.5		7
25.5-29.5		5
29.5-33.5		4
33.5-37.5		2
37.5-41.5		1
		Total = 40

Q.3] Prepare a frequency distribution table for the following data using Exclusive class method.

Use classes as 15-25, 25-35, 35-45. Total: 50.

Data:-

15 45 40 42 65 69 40 35 37 40
75 75 80 81 50 60 62 68 70 42
31 45 42 43 25 26 31 32 78 45
60 62 58 43 55 56 78 80 81 62
75 62 68 45 69 70 50 72 56 58.

→ For Exclusive class method.

class	Frequency	Count
15-25	I	1
25-35		5
35-45		10
45-55	I	6
55-65		11
65-75		8
75-85		9
		Total: 50.

Q.4] Prepare a frequency distribution table for the following data using Exclusive class method.

Use classes are : 20-40, 40-60, 60-80, 80-100

Data:-

20 41 45 24 23 45 78 87 58 65
62 22 90 91 89 75 72 70 62 63
29 42 45 51 85 65 44 27 32 38
75 92 90 47 62 61 33 47 45 60
74 76 81 80 40 50 56 23 55 60

class	Frequency	Count
20-40		9
40-60		16
60-80		16
80-100		9
		Total = 50

8] Prepare relative frequency distribution for the following data.

Wage Range(₹)	30-40	40-50	50-60	60-70	70-80	80-90
Number of Workers	10	20	40	16	8	6

→ Calculate of Relative frequency

Total number of Workers:-

$$N = 10 + 20 + 40 + 16 + 8 + 6 = 100$$

Formula : Relative frequency = $\frac{f}{N}$

Wage Range (₹)	frequency (f)	Relative frequency (f/N)
30-40	10	0.10
40-50	20	0.20
50-60	40	0.40
60-70	16	0.16
70-80	8	0.08
80-90	6	0.06
Total	$N=100$	<u>1.00</u>

$\therefore$ Total Relative frequency = 1.00.

9] Prepare relative frequency distribution for the following data:

class Intervals.	50-60	60-70	70-80	80-90	90-100	100-110
Frequency	30	25	45	15	20	40

Total frequency:

$$N = 30 + 25 + 45 + 15 + 20 + 40 = 175.$$

class Interval	Frequency (f)	Relative frequency (f/N)
50-60	30	0.1714
60-70	25	0.1429
70-80	45	0.2571
80-90	15	0.0857
90-100	20	0.1143
100-110	40	0.2286
Total	$N = 175$	$\sum f/N = 1.0000$

$\therefore$ Total Relative frequency = 1.000.

10] Prepare Cumulative frequency distributions for the following data type less than and more than.

Values	10	15	20	25	30
Frequency	3	2	4	1	5

Values	Frequency	less than (cf)	more than (cmf)
10	3	3	mf = 15
15	2	5	12
20	4	9	10
25	1	10	6
30	5	cf 15	5

$\therefore$ Total $N = 15$

$\therefore N = 15$

Less than (C.f) = 15

More than (m.f) = 15.